

Total No. of Questions : 12]

SEAT No. :

P-2968

[Total No. of Pages : 3

[6004]-951

**B.E. (Mechanical Engineering)
INDUSTRIAL ENGINEERING**

(2015 Pattern) (Semester - II) (Elective - III) (402049B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of programmable calculator is not permitted.
- 5) Assume suitable data, if necessary.

Q1) Define Industrial Engineering. Explain in brief various function of industrial Engineering. **[8]**

OR

Q2) What is productivity? Explain any four productivity improvement techniques. **[8]**

Q3) Explain Multiple activity chart and two handed process chart. **[8]**

OR

Q4) What is operational process chart? Explain with example, Method Study symbols for recording facts. **[8]**

Q5) What are the various allowances used. Why allowances are considered in the time study? **[8]**

OR

P.T.O.

Q6) Following are the element times of machine operation. The corresponding rating and relaxation allowances are given in following table [8]

Element	Observed time (min)	Performance rating (%)	Relaxation allowance (%)
1	0.15	80	13
2	0.05	85	13
3	0.55	90	10
4	1.00	95	12
5	0.1	90	13

Q7) a) Explain the functions of PPC. [8]

b) Write note on MRP, MRP-II. [8]

OR

Q8) a) Write a note on forecasting methods. [8]

b) Explain Concept of ERP with block diagram. [8]

Q9) a) Explain in detail factors affecting on selection of location for the industrial plant. [7]

b) Write a short note on : [8]

i) ABC analysis

ii) Assembly Line Balancing

OR

Q10) a) Estimate the sales forecast for the year 2000, using three month moving average and exponential smoothing forecast. Take $\alpha = 0.5$. [8]

Year	1995	1996	1997	1998	1999
Sale Rs. ($\times 10^5$)	180	168	159	170	188

b) What are different principles of material handling? Describe any four material handling equipment commonly used in industries. [7]

Q11) a) Write short note on : [7]

- i) Break even Analysis
- ii) Push and Pull systems

b) A company has given following information [8]

Data	Rs.
Selling price	10 / unit
Variable cost	2 / unit
Fixed cost	40,000/-

Determine :

- i) Profit when sales are Rs. 1,00,000/-
- ii) BEP
- iii) Sales when it is desirable to earn profit of Rs. 20,000/-

OR

Q12) a) Write a short note on (any two): [8]

- i) KRA
- ii) Cost Accounting
- iii) Net present value (NPV)

b) Explain in brief different functions of human resource department.
Comment on Key Result Area and Performance appraisal of employee.

[7]

